

Supplementary Material

Tracked but Suppressed: How LLMs Fail State Tracking

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A Dataset Details

We construct two datasets that share the same (K, N) schema and grid but differ in surface form: **Arbitrary-Single (ARB)** uses single-token English words as values, while **Semantic-Multi (SEM)** uses multi-token real category members. Both isolate position-indexed retrieval from semantic shortcuts.

Arbitrary-Single (ARB). A pool of 2,300 single-token English words organised into 46 semantic categories (e.g., *color, fruit, metal*). For each trial, K categories are sampled as keys and N values are drawn without replacement from each category’s pool. Keys and values share no inherent update semantics — there is no reason to expect any particular ordering among values of the same category. This isolates tracking from semantic shortcuts and provides the single-token values required for the mechanistic analysis (§7).

Semantic-Multi (SEM). A pool of 2,403 multi-token values across the same 46 categories, where values are real members of each category (e.g., *gemstone* → *alexandrite, amethyst, ...*). Values are multi-token and drawn from a smaller per-category pool (45–70 values), limiting feasible update counts to $N \leq 50$ for most key counts. SEM is the primary behavioural dataset because the realistic key–value semantics rule out trivial template-matching strategies.

Grid coverage. On the unified grid $K \in \{2, 3, 5, 7, 10, 15, 20, 25, 30\}$, $N \in \{5, 7, 10, 15, 20, 30, 50, 75, 100\}$ (§3), cells where the prompt exceeds a model’s context limit are excluded; cells where the dataset pool is insufficient are also excluded (relevant for SEM at high N). This yields up to 79 feasible cells per model on ARB and up to 63 on SEM.

B Prompts, Matching Rule, and Seeds

This appendix documents the exact text used to construct the inference prompts (Figure 1), the four format variants used in §5, the string-matching rule used to score model responses, and the deterministic seeding scheme.

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Naming convention. The main text refers to the second query type as *CVQ* (current-value query), reflecting its meaning. The actual prompt text asks for the *last* value of the target key, which is operationally equivalent when the stream is the complete record of updates.

System prompt and queries

System prompt:

You are a precise data extraction tool. Output ONLY a single word — the exact value requested. No other text, no explanation, no punctuation.

Queries (rendered with the target key substituted in):

FVQ: *What was the first value of [TARGET]?*

CVQ: *What was the last value of [TARGET]?*

IVQ: *What was the [K]-th value of [TARGET]?*

Figure 1: Inference-time system prompt and per-query phrasing used in all behavioural sweeps reported in the paper. “CVQ” in the main text corresponds to the literal “last value of” query above.

Stream formats. The four formats compared in §5 differ only in how the stream lines are rendered and which user-turn preamble introduces them. Figure 2 shows a small ($K=2, N=3$) example in each format.

Chat-template rendering. For instruction-tuned models the user/system messages are rendered through the model’s tokenizer-default chat template; for base models without a chat template we use a completion-style few-shot wrapper described in the released code.

Matching rule. A model response is counted as correct if, after lowercasing and stripping whitespace, the expected value either equals the response, is contained in the response, or is a prefix of the response. This prefix-tolerant case-insensitive rule accommodates multi-token semantic values where models occasionally emit a trailing qualifier; spot-checks confirm it does not credit lexically unrelated strings.

Seeds. For each $(K, N, \text{query type}, \text{trial index})$ tuple we derive a deterministic seed by hashing the tuple. This ensures the identical stream is presented across base-vs-intervention

Plain (flat_nolabel; default in §4) <i>Read the following key-value stream. Each key appears multiple times as it gets updated. Count each occurrence of a key as one update.</i> <pre>color: red size: large color: blue size: small color: green size: medium</pre> <i>What was the 3rd value of color? Answer with ONLY the exact value. No explanation.</i>	Labeled (flat_verbose) <i>Read the following key-value stream. Each key is updated multiple times.</i> <pre>color (update 1): red size (update 1): large color (update 2): blue size (update 2): small color (update 3): green size (update 3): medium</pre> <i>What was the value of color in Update 3? Answer with ONLY the exact value. No explanation.</i>
Block (block) <i>Read the following key-value stream. Each key is updated multiple times, grouped by update round.</i> <pre>[Update 1] color: red size: large [Update 2] color: blue size: small [Update 3] color: green size: medium</pre> <i>What was the value of color in Update 3? Answer with ONLY the exact value. No explanation.</i>	Landmark (landmark) <i>Read the following key-value stream. Each key is updated multiple times, grouped by update round. The ‘—’ marker separates consecutive update rounds.</i> <pre>color: red size: large -- color: blue size: small -- color: green size: medium</pre> <i>What was the value of color in Update 3? Answer with ONLY the exact value. No explanation.</i>

Figure 2: The four prompt formats compared in §5, illustrated on a $K=2, N=3$ instance. Same underlying stream content; only the surface structure changes. The “Plain” format is the default for all main-body figures and tables (§4).

comparisons at the same cell and trial index, so the two conditions are scored on identical inputs.

C Model Details

Table 1 lists every model evaluated in this paper. The first block is the main open-weight set whose behaviour is analysed in the main text; the second block is three smaller auxiliary models reported in Appendix D for completeness; the third block is the proprietary frontier models. IT = instruction-tuned.

Model licenses. All open-weight models are used under the license listed on each model’s HuggingFace page (verified individually). Proprietary models are accessed through their official APIs under each provider’s terms of service.

- Qwen/Qwen2.5-0.5B-Instruct (HuggingFace): **Apache 2.0.**
- Qwen/Qwen2.5-1.5B-Instruct (HuggingFace): **Apache 2.0.**
- Qwen/Qwen2.5-3B-Instruct (HuggingFace): **Qwen Research License** (<https://huggingface.co/Qwen/Qwen2.5-3B-Instruct/blob/main/LICENSE>).
- Qwen/Qwen2.5-3B (HuggingFace): **Qwen Research License.**
- Qwen/Qwen3.5-0.8B (HuggingFace): **Apache 2.0.**
- Qwen/Qwen3.5-2B (HuggingFace): **Apache 2.0.**
- Qwen/Qwen3.5-4B (HuggingFace): **Apache 2.0.**
- Qwen/Qwen3.5-9B (HuggingFace): **Apache 2.0.**
- google/gemma-3-270m-it (HuggingFace): **Gemma Terms of Use** (<https://ai.google.dev/gemma/terms>).
- google/gemma-3-1b-it (HuggingFace): **Gemma Terms of Use.**
- google/gemma-3-4b-it (HuggingFace): **Gemma Terms of Use.**
- TinyLlama/TinyLlama-1.1B-Chat-v1.0 (HuggingFace): **Apache 2.0.**
- stabilityai/stablelm-2-1_6b-chat (HuggingFace): **Stability AI Non-Commercial Research Community License** (https://huggingface.co/stabilityai/stablelm-2-1_6b-chat/blob/main/LICENSE.md).
- EleutherAI/pythia-410m (HuggingFace): **Apache 2.0.**

Model	Family	Params	IT?	Ctx
<i>Main open-weight</i>				
Qwen2.5-0.5B-Instruct	Qwen2.5	0.5B	✓	32K
Qwen2.5-1.5B-Instruct	Qwen2.5	1.5B	✓	32K
Qwen2.5-3B-Instruct	Qwen2.5	3B	✓	32K
Qwen2.5-3B	Qwen2.5	3B	×	32K
Qwen3.5-0.8B	Qwen3.5	0.8B	✓	256K
Qwen3.5-2B	Qwen3.5	2B	✓	256K
Qwen3.5-4B	Qwen3.5	4B	✓	256K
Qwen3.5-9B	Qwen3.5	9B	✓	256K
Gemma-3-270m-it	Gemma-3	270M	✓	32K
Gemma-3-1b-it	Gemma-3	1B	✓	32K
Gemma-3-4b-it	Gemma-3	4B	✓	128K
<i>Auxiliary (appendix only)</i>				
TinyLlama-1.1B	LLaMA	1.1B	✓	2K
StableLM-2-1.6B	StableLM	1.6B	✓	4K
Pythia-410M	Pythia	410M	×	2K
<i>Proprietary frontier</i>				
GPT-4.1	OpenAI	—	—	1M
GPT-4.1-mini	OpenAI	—	—	1M
Claude-4.5-Haiku	Anthropic	—	—	200K
Claude-4.5-Sonnet	Anthropic	—	—	1M
Gemini-2.5-Flash	Google	—	—	1M
Gemini-2.5-Pro	Google	—	—	1M

Table 1: All 20 models evaluated, grouped by role: 11 main open-weight models that drive the main analyses; 3 smaller auxiliary models reported only in Appendix D; and 6 proprietary frontier models. **Ctx** is the advertised input context window (HF `max_position_embeddings` for open-weight; provider documentation for proprietary; see Appendix D).

- HuggingFaceTB/SmolLM2-1.7B-Instruct (HuggingFace): **Apache 2.0**.
- HuggingFaceTB/SmolLM3-3B (HuggingFace): **Apache 2.0**.
- **GPT-4.1, GPT-4.1-mini** (OpenAI API): OpenAI Terms of Service (<https://openai.com/policies/terms-of-use/>).
- **Claude-4.5-Haiku, Claude-4.5-Sonnet** (Anthropic API): Anthropic Commercial Terms of Service (<https://www.anthropic.com/legal/commercial-terms>).
- **Gemini-2.5-Flash, Gemini-2.5-Pro** (Google API): Gemini API Additional Terms of Service (<https://ai.google.dev/gemini-api/terms>).

The LoRA adapter we release is distributed under the **Qwen Research License**, matching the license of its base model (Qwen/Qwen2.5-3B-Instruct).

D Full Model Results

Context-usage measurement (Table 1). The % Ctx column of Table 1 reports the fraction of each model’s advertised context window used by one canonical Semantic-Multi prompt at the row’s representative cell. Open-weight rows use the $K=5, N=30$ prompt (1,029 tokens); proprietary rows use the $K=20, N=50$ prompt (6,565 tokens). All counts use `tiktoken o200k_base` (the GPT-4o / GPT-4.1 tokeniser) for uniform cross-provider comparability; on-row tokenisation with each model’s actual tokeniser changes individual counts by under $\pm 5\%$ for English text and does not move any row above 5% of its window.

Context-window sources. Open-weight windows are read from each model’s `config.json` on the HuggingFace Hub (`max_position_embeddings` in `text_config` for multi-modal models): Qwen2.5-3B-Instruct 32,768; Qwen3.5-2B, -4B, -9B 262,144 each; Gemma-3-1b-it 32,768; Gemma-3-4b-it 131,072 (rope-scaling factor 8 over the 16,384-token base). Proprietary windows are advertised input limits from provider documentation: GPT-4.1 and GPT-4.1-mini 1,000,000 tokens (<https://openai.com/index/gpt-4-1/>); Claude-4.5-Haiku 200,000 tokens; Claude-4.5-Sonnet 1,000,000 tokens via the long-context option (<https://docs.anthropic.com/en/docs/about-claude/models/overview>); Gemini-2.5-Flash 1,048,576 and Gemini-2.5-Pro 1,000,000 (<https://ai.google.dev/gemini-api/docs/models>).

E Format Intervention Data

This appendix supports two claims made in the main text: the format-intervention results in §5.1 (Table 2, in §5.1) and the multi-model intermediate-position retrieval claim in §4.

Per-model patterns under Labeled and Landmark. The full 4-format $\times$ 11-model results in Table 2 (§5.1) reveal two distinct partial-recovery patterns for the intermediate formats.

Labeled (per-entry numbering, no grouping) tends to push FVQ and CVQ together rather than restoring CVQ to ceiling. In Qwen2.5-3B-Instruct, FVQ drops from 0.49 (Plain) to 0.34 while CVQ rises only to 0.39 — both land near

0.4, shifting the failure rather than removing it. Gemma-3-4b-it shows the same collapse: FVQ falls from 0.92 to 0.39 and CVQ stays near 0.47. The proprietary set tolerates Labeled better, suggesting that per-entry indexing is a noisy cue smaller open-weight models cannot disambiguate from update content.

Landmark (round separators, no numbers) preserves FVQ but fails to recover CVQ in the open-weight set: Gemma-3-4b-it stays at $FVQ \approx 0.97$ / $CVQ \approx 0.00$, Qwen3.5-2B at $FVQ \approx 0.80$ / $CVQ \approx 0.07$. Proprietary models recover more cleanly under Landmark, but the gap remains positive there.

Read together, Labeled and Landmark localise Block’s two effective ingredients: round structure provides salient boundaries for the lookup region (without which CVQ stays floored), while explicit round numbers anchor that structure to the query (without which open-weight models fall back to Plain-format behaviour). Block combines both.

Wilson 95% half-widths. Table 2 reports the same per-(model, format) FVQ and CVQ accuracies as Table 2 with Wilson 95% half-widths (Wilson 1927; up to 200 trials per cell). Half-widths are ≤ 0.07 throughout; the Plain-to-Block CVQ recoveries cited in §5.1 (e.g. Qwen3.5-2B $0.01 \rightarrow 0.98$, Qwen3.5-4B $0.00 \rightarrow 1.00$, Gemma-3-4b-it $0.01 \rightarrow 0.89$) exceed combined-CI overlap by a wide margin. The Labeled equalisation cases discussed above (Qwen2.5-3B-Instruct $FVQ\ 0.49 \rightarrow 0.34$, $CVQ\ 0.23 \rightarrow 0.39$; Gemma-3-4b-it $FVQ\ 0.92 \rightarrow 0.39$, $CVQ\ 0.01 \rightarrow 0.47$) show CIs that exclude the corresponding Plain values in every case.

Multi-model per-position retrieval under the default Plain format. Figure 3 shows per-position retrieval accuracy at $K=10$, $N=50$ on Arbitrary-Single under the default Plain format, separately for the five open-weight models in the ucurve sweep and the six proprietary frontier models. All five open-weight models collapse below 5% accuracy within two positions of the stream start. The proprietary panel reveals three regimes: three models (GPT-4.1, GPT-4.1-mini, Claude-4.5-Haiku) collapse like the open-weight set; Claude-4.5-Sonnet and Gemini-2.5-Flash show partial recovery; Gemini-2.5-Pro alone maintains near-ceiling accuracy across all queried positions.

Block format restores per-position retrieval. Figure 4 compares per-position accuracy under Plain vs Block format on the same five open-weight models. Plain collapses below 5% accuracy by position 2 across every model; Block restores accuracy to 0.70–0.99 at every queried position. This substantiates the §5.2 close claim that the format intervention recovers intermediate-position retrieval, not only the FVQ–CVQ gap.

Block 2×2 decomposition. The Block format combines two design properties: round-grouping (entries clustered by update round) and round-numbering (headers labelling each round). The Labeled and Landmark formats separate these: Labeled uses per-entry numbering without grouping, Landmark uses grouping without numbering. We decompose Block’s effect across 36 (model, K , N) cells from the proprietary ucurve sweep, bootstrap 95% CIs ($n_{boot}=2000$, seed

17).

Reading the decomposition (Table 3). For CVQ, both ingredients help independently (+10.6 pp from numbering, +13.6 pp from grouping; both CIs exclude zero). Block’s combined gain (+14.3 pp) is less than the sum (super-additivity -0.10 , CI excludes zero): the two ingredients partially overlap. For FVQ, numbering alone is null and grouping alone *hurts* (-14.2 pp); Block recovers because numbering compensates the grouping cost. Block’s ability to lift CVQ *without* sacrificing FVQ is therefore not attributable to either property alone — it depends on the combination.

F Dense Intermediate-Position Retrieval — Full Sweep

Figure 3 in §5 shows position-by-position retrieval accuracy at a single cell ($K=5$, $N=10$). To substantiate the broader claim that the failure is a general loss of position-indexed retrieval, we report the same measurement across all six cells we tested. Figure 5 shows the per-position curves; Table 4 summarises first / intermediate / last position accuracy per cell. Across all six (K , N) cells, baseline intermediate-position accuracy stays at or below 25% and the LoRA-tuned model recovers near-ceiling accuracy at every position queried.

G LoRA Training Details

This appendix documents the full configuration of the LoRA intervention reported in §5.2. All scripts, training data, and checkpoint selection logic are in the released codebase.

Adapter and optimiser. The adapter is fitted with PEFT (Hu et al. 2022) on top of the frozen base model. Table 5 lists the hyperparameters; LoRA matrices are inserted into the four attention projections $\{Q, K, V, O\}$ of every transformer layer, and MLP and embedding weights are left unmodified.

Training data composition. Each training example is one (K , N) stream with a single query appended. The 18,000 training examples are split 40% FVQ, 40% CVQ, 20% intermediate-position queries (IVQ). Including IVQ during training removes a possible shortcut: a model that simply amplified a recency-attending circuit could pass FVQ plus CVQ supervision while leaving intermediate positions unchanged; IVQ supervision blocks that path. Training examples are sampled uniformly across the 16 training cells, yielding $\sim 1,125$ examples per cell.

Training and held-out cell split. The training and evaluation cells are disjoint by design. Table 6 lists both grids; held-out cells extend roughly $10\times$ beyond training in stream length ($K \times N$ from ≤ 200 at training to $\leq 2,250$ at evaluation).

Category split. Of the 46 semantic categories in the Arbitrary-Single pool (Appendix A), 35 are used during training and 11 are held out for evaluation; no category appears in both splits (Table 7).

Model	Plain	Labeled	Block	Landmark
<i>Open-weight models</i>				
Qwen2.5-3B-Instruct	0.49 / 0.23 $\pm 0.07 / \pm 0.06$	0.34 / 0.39 $\pm 0.06 / \pm 0.07$	0.84 / 0.92 $\pm 0.05 / \pm 0.04$	0.71 / 0.27 $\pm 0.06 / \pm 0.06$
Qwen3.5-2B	0.95 / 0.01 $\pm 0.03 / \pm 0.02$	0.85 / 0.93 $\pm 0.05 / \pm 0.04$	1.00 / 0.98 $\pm 0.01 / \pm 0.02$	0.80 / 0.07 $\pm 0.06 / \pm 0.03$
Qwen3.5-4B	0.99 / 0.00 $\pm 0.02 / \pm 0.01$	0.93 / 0.97 $\pm 0.04 / \pm 0.02$	1.00 / 1.00 $\pm 0.02 / \pm 0.02$	0.99 / 0.47 $\pm 0.01 / \pm 0.07$
Gemma-3-4b-it	0.92 / 0.01 $\pm 0.04 / \pm 0.02$	0.39 / 0.47 $\pm 0.07 / \pm 0.07$	0.99 / 0.89 $\pm 0.02 / \pm 0.04$	0.92 / 0.00 $\pm 0.04 / \pm 0.01$
<i>Proprietary models</i>				
GPT-4.1	1.00 / 0.69 $\pm 0.01 / \pm 0.06$	1.00 / 0.96 $\pm 0.02 / \pm 0.05$	1.00 / 0.98 $\pm 0.03 / \pm 0.04$	0.99 / 0.99 $\pm 0.01 / \pm 0.01$
GPT-4.1-mini	1.00 / 0.77 $\pm 0.01 / \pm 0.06$	1.00 / 0.94 $\pm 0.02 / \pm 0.05$	0.99 / 0.99 $\pm 0.03 / \pm 0.03$	0.78 / 1.00 $\pm 0.06 / \pm 0.01$
Claude-4.5-Haiku	1.00 / 0.43 $\pm 0.01 / \pm 0.07$	1.00 / 0.95 $\pm 0.02 / \pm 0.04$	1.00 / 0.99 $\pm 0.01 / \pm 0.02$	0.91 / 0.90 $\pm 0.04 / \pm 0.04$
Claude-4.5-Sonnet	0.99 / 0.80 $\pm 0.01 / \pm 0.06$	1.00 / 0.81 $\pm 0.01 / \pm 0.05$	1.00 / 1.00 $\pm 0.04 / \pm 0.04$	0.96 / 0.99 $\pm 0.03 / \pm 0.01$
Gemini-2.5-Flash	1.00 / 0.76 $\pm 0.01 / \pm 0.06$	1.00 / 0.98 $\pm 0.03 / \pm 0.04$	1.00 / 1.00 $\pm 0.03 / \pm 0.03$	1.00 / 0.95 $\pm 0.02 / \pm 0.05$
Gemini-2.5-Pro	1.00 / 1.00 $\pm 0.03 / \pm 0.03$	1.00 / 1.00 $\pm 0.04 / \pm 0.04$	1.00 / 1.00 $\pm 0.04 / \pm 0.04$	0.54 / 0.99 $\pm 0.07 / \pm 0.01$

Table 2: Format-intervention results with Wilson 95% half-widths, $K=10$, $N=50$. Each cell shows FVQ / CVQ on the first line and the matching $\pm$ half-widths on the second. Same data as Table 2.

CVQ contrast	Δ [95% CI]
Numbering (Labeled – Plain)	+0.106 [+0.065, +0.153]
Grouping (Landmark – Plain)	+0.136 [+0.099, +0.177]
Combined (Block – Plain)	+0.143 [+0.085, +0.213]
Super-additivity	−0.100 [−0.143, −0.062]
FVQ contrast	Δ [95% CI]
Numbering (Labeled – Plain)	+0.002 [−0.000, +0.005]
Grouping (Landmark – Plain)	−0.142 [−0.210, −0.086]
Combined (Block – Plain)	−0.001 [−0.005, +0.001]
Super-additivity	+0.141 [+0.085, +0.213]

Table 3: Block 2×2 decomposition. Both numbering and grouping contribute to CVQ recovery, but their combination undershoots the sum (sub-additive). For FVQ, grouping alone hurts (−14.2 pp) and numbering rescues; Block’s no-cost FVQ behaviour depends on the combination.

Compute Budget and Software Stack. Open-weight model evaluation (14-model behavioural sweep, format-intervention sweep, intermediate-position sweep, training-dynamics across 79 SmolLM2/3 checkpoints, mechanism probing) and the main Qwen2.5-3B-Instruct LoRA training ran on a single NVIDIA L40S GPU node, totalling roughly 120 GPU-hours wall-clock. The Gemma-3-4b-it family-control LoRA and the cross-family mechanism analyses ran on a SageMaker NVIDIA L4 node. Proprietary-model evaluation is via official APIs and uses no local GPU. All decoding is greedy (temperature 0) with maximum new tokens 32.

Hardware:

- NVIDIA L40S (48 GB VRAM): main open-weight evaluation, Qwen2.5-3B-Instruct LoRA training. ~ 120 GPU-hours total.
- NVIDIA L4 (23 GB VRAM, Ubuntu 24.04 LTS, driver 595.58.03): Gemma family-control LoRA, cross-family mechanism analyses.
- Proprietary APIs (no local GPU): GPT-4.1, GPT-4.1-mini, Claude-4.5-Haiku, Claude-4.5-Sonnet, Gemini-2.5-Flash, Gemini-2.5-Pro.

Software:

- Python 3.12.13; PyTorch 2.11.0 (CUDA 13.0, cuDNN 9.19).
- transformers 4.57.6, tokenizers 0.22.2, safetensors 0.7.0, sentencepiece 0.2.1, huggingface-hub 0.36.2.
- LoRA training: peft 0.19.1, trl 1.4.0 (SFTTrainer), datasets 4.8.5, accelerate 1.13.0, bitsandbytes 0.49.2.
- Batched inference: vllm 0.21.0.
- Mechanistic analysis: transformer_lens 3.2.1.
- Probing + statistics: scikit-learn 1.7.2, numpy 2.3.5, pandas 3.0.3.
- Plotting: matplotlib 3.10.9.

Pre-registered controls. We pre-registered two controls before running the main experiment. The **family control** trains the identical adapter configuration (rank, learning rate, schedule, data mix) on Gemma-3-4b-it; success here indicates the recovery is architecture-agnostic rather than specific to Qwen2.5. Results are reported in App G.1 (28/28 ARB

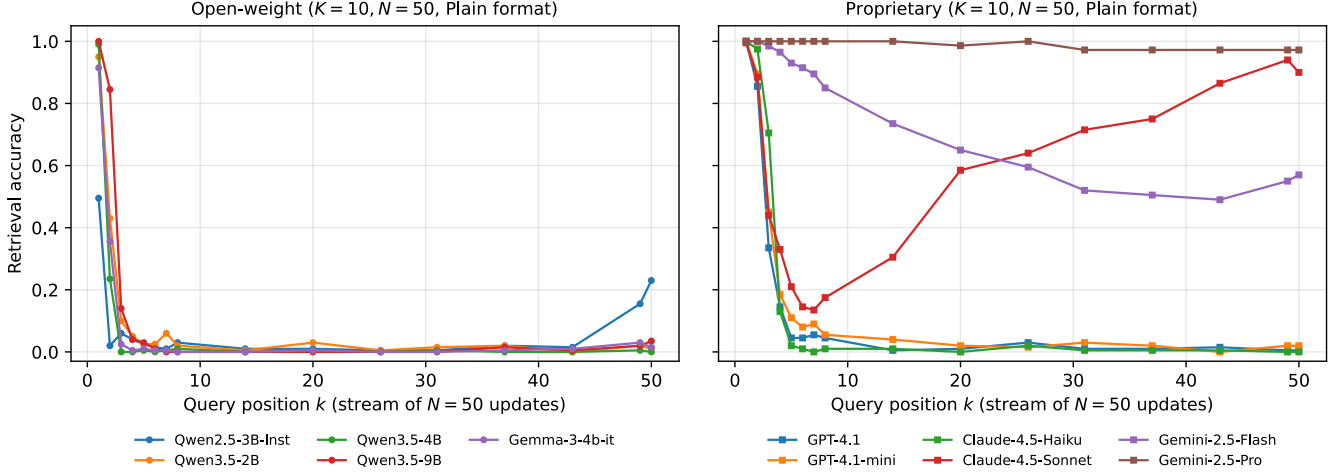


Figure 3: Per-position retrieval accuracy on Arbitrary-Single at $K=10$, $N=50$ under the default Plain format (ucurve sweep; `flat_nolabel` condition). Each line is one model. **Left:** five open-weight models, all collapsing uniformly to near-floor by position 2. **Right:** six proprietary models; most collapse but Gemini-2.5-Pro maintains near-ceiling accuracy throughout, and Claude-4.5-Sonnet shows U-shaped recovery toward the end of the stream.

(K, N)	Base			+ LoRA		
	Pos. 1	Interm. min-max	Last	Pos. 1	Interm. min-max	Last
(5, 10)	80%	0%–25%	0%	100%	70%–100%	60%
(10, 20)	25%	0%–20%	5%	100%	5%–100%	35%
(10, 30)	65%	0%–10%	5%	100%	0%–100%	0%
(15, 30)	20%	0%–5%	0%	100%	0%–90%	0%
(20, 30)	30%	0%–5%	0%	90%	0%–85%	0%
(25, 50)	5%	0%–5%	0%	90%	0%–70%	0%

Table 4: Per-cell summary of dense intermediate-position retrieval accuracy on Arbitrary-Single. “Interm. min-max” is the range across all intermediate positions ($1 < k < N$) at that cell. The base model loses accuracy at every intermediate position across every cell; the LoRA-tuned model holds near ceiling.

Setting	Value
Base model	Qwen2.5-3B-Instruct
LoRA rank	16
LoRA α	32
LoRA dropout	0.05
Target modules	q/k/v/o_proj
Trainable parameters	7.4M (0.24% of model)
Learning rate	2×10^{-4}
LR schedule	cosine, 100-step linear warm-up
Optimiser	paged AdamW (8-bit)
Epochs	2
Batch size	8 ($\times 8$ accum = eff. 64)
Max sequence length	2,048
Mixed precision	bfloat16
Training examples	18k (+2k val, different seeds)
Checkpoint selected	best val loss (step 564, end of ep. 2)

Table 5: LoRA training hyperparameters.

Split	Cells (K, N)
Training (16)	$K \in \{2, 3, 5, 10\}, N \in \{5, 10, 15, 20\}$
Held-out (28)	$K=7, N \in \{30, 50, 75\}$ $K=10, N \in \{30, 50, 75\}$ $K=15, N \in \{15, 20, 30, 50, 75\}$ $K=20, N \in \{15, 20, 30, 50, 75\}$ $K=25, N \in \{10, 15, 20, 30, 50, 75\}$ $K=30, N \in \{10, 15, 20, 30, 50, 75\}$

Table 6: Training and held-out cells for the LoRA intervention. Training cells cover only low (K, N) ; held-out cells extend to $K=30$ and $N=75$, well outside the training distribution in both dimensions.

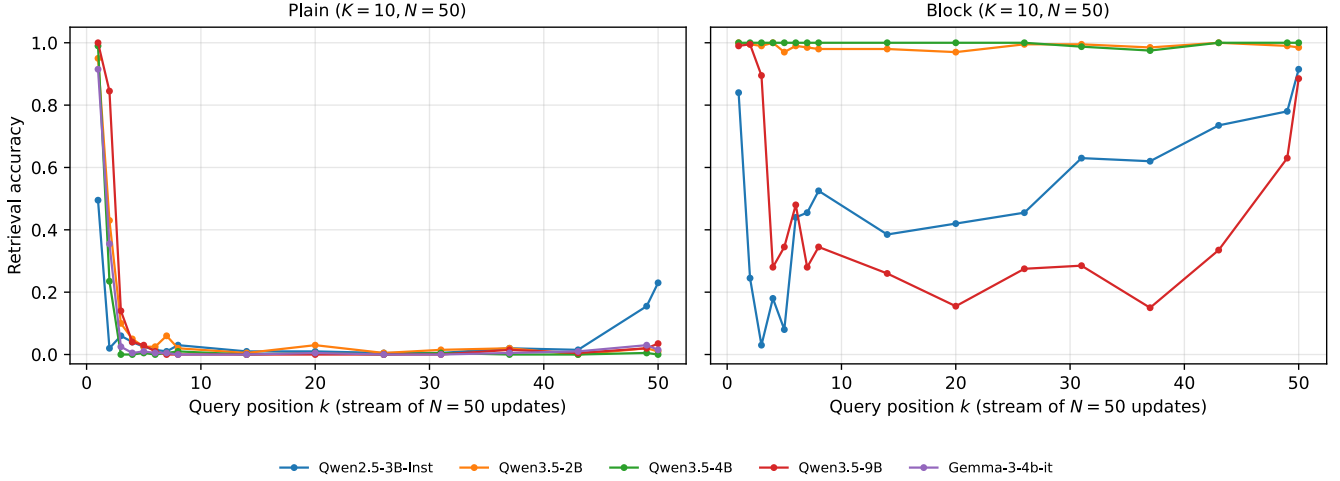


Figure 4: Per-position retrieval at $K=10$, $N=50$ on Arbitrary-Single, five open-weight models, Plain (left) vs Block (right) format. Block restores intermediate-position accuracy across every model, paralleling the LoRA recovery shown for Qwen2.5-3B-Instruct in Figure 3.

Training (35)	Held-out (11)
<i>visual art, tools, landform, musical instrument, gemstone, fabric, tree species, cheese variety, architectural style, cloud formation, bird species, culinary herb, flower species, wine variety, dance style, pasta shape, literary genre, cooking method, mathematical concept, weather phenomenon, ocean current, mineral type, coffee variety, telescope type, martial art, sea creature, psychology term, chemical element, dinosaur genus, programming language, ancient civilization, bridge type, photography technique, boat type, cartoon character</i>	<i>stadium name, surgical procedure, constellation, spice blend, guitar type, hat style, painting medium, volcano name, fruit variety, sword type, board game</i>

Table 7: LoRA category split. 35 categories used during training, 11 held out for evaluation; no overlap.

cells fixed, 16/16 SEM OOD cells fixed). The **negative control** fine-tunes the same Qwen2.5-3B-Instruct with identical hyperparameters (rank-16 attention-only LoRA, two epochs, matched batch and learning rate) on GSM8K — chain-of-thought arithmetic word problems, $\sim 6.7k$ training examples — and then evaluates on the FVQ/CVQ grid. The adapter learned its training task: GSM8K end-to-end accuracy rose from 16% to 70% on a 250-problem test split. It did not close the FVQ–CVQ gap. Table 8 reports five held-out cells (100 trials each, Wilson 95% CIs); FVQ improved modestly while CVQ *decreased* on every cell relative to baseline (post-LoRA CVQ CIs do not overlap with base). No cell reached the pre-registered fix threshold (FVQ ≥ 0.65 and CVQ ≥ 0.65); the task-specific LoRA in §5.2 reaches it on 28/28 cells. The recovery is therefore task-specific, not a general consequence of LoRA fine-tuning.

Full held-out results. Table 9 reports the full FVQ and CVQ accuracies that underlie the compact gap-only Table 3 in §5.2.

G.1 Gemma-3-4b-it family control

The family-control LoRA fits an identical adapter to Gemma-3-4b-it (grouped-query attention, 34 layers $\times$ 8 query heads with KV grouping). The training recipe matches the Qwen run: rank 16, LoRA $\alpha = 32$, attention-only $\{Q, K, V, O\}$, dropout 0.05, lr 2×10^{-4} cosine with 100-step warm-up, effective batch 64, bf16 mixed precision, gradient checkpointing. The adapter contributes 8.9M trainable parameters (0.23% of model weights). Training data is regenerated under the same $K \in \{2, 3, 5, 10\}$, $N \in \{5, 10, 15, 20\}$ grid and the same 40/40/20 FVQ/CVQ/IVQ mix; the concrete training examples differ from the Qwen run (deterministic seeding was introduced between runs).

Training runs for 400 steps (1.42 epochs), reaching eval accuracy 98.3% with eval loss still improving. The model attaches to text-only inputs via the `Gemma3ForCausalLM` class with `sdpa` attention; defaults load Gemma’s multi-modal processor which is incompatible with our text-only data format.

Tables 10 and 11 report the held-out cell results. Across all 28 held-out Arbitrary-Single cells, both FVQ and CVQ accuracies reach 88–100% post-adapter (mean FVQ 99.4%, mean CVQ 97.0%). Across all 16 Semantic-Multi out-of-

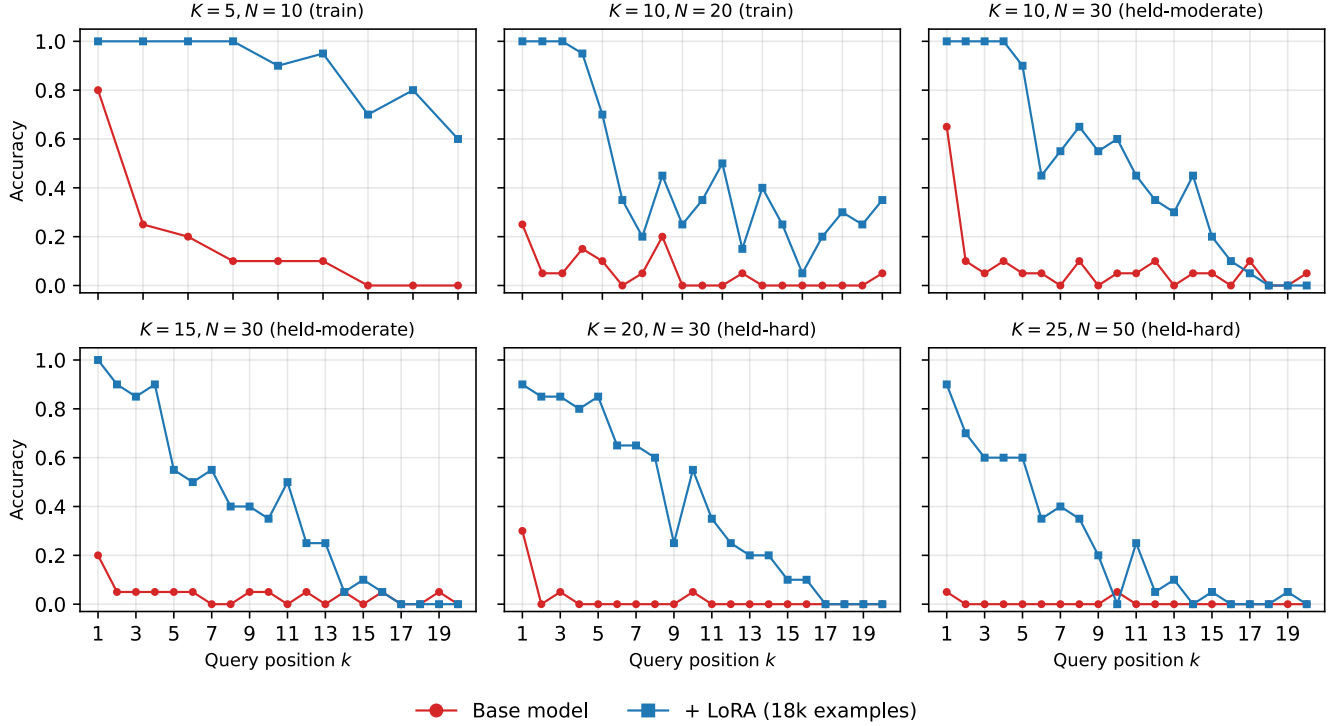


Figure 5: Per-position retrieval accuracy at all six tested (K, N) cells on Arbitrary-Single, base model vs LoRA-tuned model, for positions $k \in [1, N - 1]$ ($k=N$ is omitted because the experiment script substitutes a CVQ “last value” query at that position). Group labels (train, held-moderate, held-hard) indicate whether the cell was included in the LoRA training grid.

Cell	Base FVQ/CVQ	Arith-LoRA
$K=10, N=50$	0.42/0.50	0.48/0.26
$K=15, N=20$	0.32/0.58	0.49/0.33
$K=20, N=30$	0.21/0.51	0.32/0.29
$K=25, N=75$	0.06/0.60	0.11/0.19
$K=30, N=75$	0.01/0.56	0.07/0.26

Table 8: Arithmetic negative-control LoRA on Qwen2.5-3B-Instruct, held-out FVQ/CVQ cells (100 trials each). Adapter learned arithmetic (GSM8K 16% $\rightarrow$ 70%) but did not recover position-indexed retrieval.

(K, N)	Base			+ LoRA		
	FVQ	CVQ	gap	FVQ	CVQ	gap
(7, 30)	50%	38%	+12%	100%	100%	+0%
(7, 50)	52%	46%	+6%	100%	96%	+4%
(10, 30)	41%	46%	−5%	100%	100%	+0%
(10, 50)	42%	50%	−8%	100%	100%	+0%
(15, 20)	32%	58%	−26%	100%	98%	+2%
(15, 30)	35%	60%	−25%	100%	98%	+2%
(15, 50)	18%	59%	−41%	100%	100%	+0%
(20, 30)	21%	51%	−30%	100%	98%	+2%
(20, 50)	8%	62%	−54%	100%	96%	+4%
(20, 75)	6%	56%	−50%	100%	96%	+4%
(25, 30)	15%	59%	−44%	100%	98%	+2%
(25, 50)	5%	43%	−38%	98%	96%	+2%
(25, 75)	6%	60%	−54%	100%	93%	+7%
(30, 30)	12%	60%	−48%	100%	98%	+2%
(30, 50)	4%	54%	−50%	100%	94%	+6%
(30, 75)	1%	56%	−55%	99%	92%	+6%

Table 9: Full held-out Arbitrary-Single results for the LoRA on Qwen2.5-3B-Instruct. Same 16 cells as Table 3; FVQ and CVQ separately with the gap, before and after the adapter.

distribution cells, FVQ stays at 100% and CVQ reaches 83–100% (mean 96.8%). Figure 6 shows the underlying baseline failure profile across the full grid.

H Training Dynamics Data

This appendix lists the per-checkpoint mean gap (and component accuracies) that underlie Figure 4 in §6.

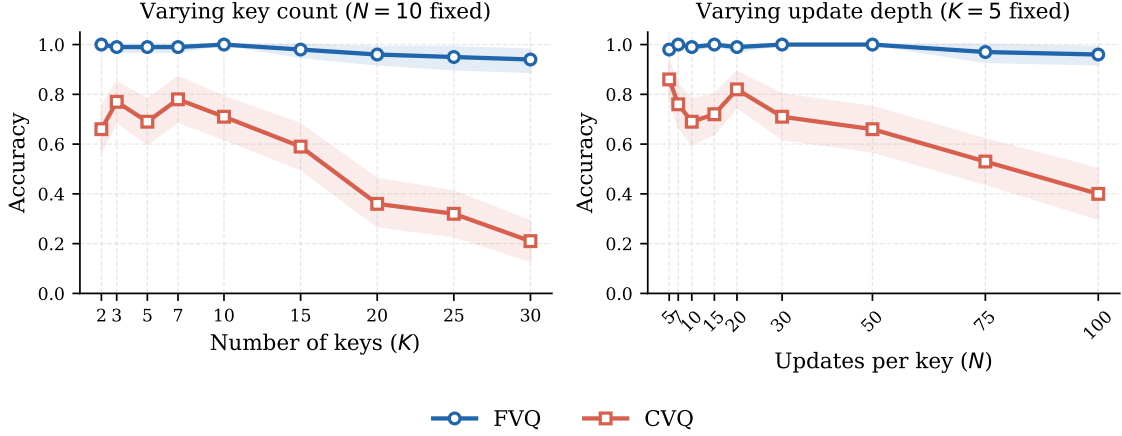


Figure 6: Baseline FVQ and CVQ accuracy on Gemma-3-4b-it across the unified grid. Clean primacy regime ($FVQ \gg CVQ$) at every cell, in contrast to Qwen2.5-3B-Instruct which crosses into the reversal regime at high stream load. The LoRA recovery in Tables 10 and 11 closes this gap on all 28 held-out cells.

Method. Each checkpoint is evaluated on Arbitrary-Single with the same query format and matching rule used in the main paper (Appendix B), at $K \in \{2, 3, 5, 7\}$, $N \in \{2, 3, 5, 7, 10, 15, 20, 30\}$, 100 trials per cell. The completion-format prompt is used at every checkpoint to ensure comparability across pretraining and post-training stages; the chat-template prompt returns near-zero accuracy on early instruction-tuned SmolLM3 checkpoints (it-mid-training under chat_1024 gives mean $FVQ = \text{mean } CVQ = 0.001$), so we report it only as a robustness comparison (Table 15). For figures that partition by load, the 32 cells split into three exhaustive bins: low ($K \leq 3$, any N ; 16 cells), medium ($K \geq 5$, $N < 20$; 12 cells), and high ($K \geq 5$, $N \geq 20$; 4 cells).

SmolLM2-1.7B trajectory. Figure 7 reports the SmolLM2-1.7B trajectory under the same three-bin partition used for SmolLM3 in §6. The pattern matches SmolLM3 with a cleaner separation: low-load cells stay positive across the published four-stage curriculum (Ben Allal et al. 2025), medium-load cells hover near zero, and high-load cells stay in the reversal direction throughout pretraining and into the released final model. Table 13 reports the per-checkpoint mean FVQ, CVQ, and gap at every released checkpoint. Aggregated across all 32 cells the gap is positive at every checkpoint, with a non-monotonic curriculum trajectory — rising through early pretraining, peaking around step 625k, partially collapsing, then stabilising near the final value.

SmolLM3-3B trajectory. Table 14 reports the same statistics for SmolLM3-3B across 32 pretraining checkpoints (the three documented stages) and the four post-training stages plus the released final model. The gap is positive at every checkpoint.

Prompt-format robustness check. For the four SmolLM3 post-training stages plus the released model we additionally ran the chat-template prompt (Table 15). Magnitudes differ between prompt formats but the sign and ordering of the gap are preserved at every stage where the chat template itself does not break the model (at it-mid-training the chat template gives near-zero accuracy on both queries, so the gap is undefined and we omit it from Figure 4).

I Cross-Family Mechanism Details (Gemma-3-4b-it)

This appendix collects the post-LoRA mechanism results on Gemma-3-4b-it underlying the cross-family replication claim in §7. Both analyses use the same merged-adapter model evaluated in App G.1; the operating point is $K=2$, $N=5$ for probing and three cells ($K=2$, $N \in \{5, 10, 50\}$) for logit lens, mirroring the Qwen analysis.

Probing classifier. Linear classifiers (sklearn logistic regression, 5-fold CV, balanced class weights) are fitted on the residual-stream activation at each of Gemma’s 34 layers, separately for three labels: *condition discrimination* (FVQ vs CVQ query), *FVQ correctness* (will the model answer FVQ correctly?), and *CVQ correctness* (will the model answer the current-value query correctly?). Table 16 reports baseline-to-LoRA Δ at the eight late layers L26–L33 with parametric 95% CIs (mean $\pm 1.96 \cdot \sigma / \sqrt{5}$, pooled variance for Δ).

Logit lens (Table 17). Per-trial residual-stream activations at the final layer (L33) are projected through Gemma’s unembedding matrix to recover layer-level token distributions; we report $\Pr(v_{\text{first}})$ for FVQ and $\Pr(v_{\text{last}})$ for CVQ, averaged over 100 trials per cell with nonparametric bootstrap 95% CIs ($n_{\text{boot}} = 2000$, seed 17).

(K, N)	Base			+ LoRA		
	FVQ	CVQ	gap	FVQ	CVQ	gap
(7, 30)	95%	52%	+43%	100%	100%	+0%
(7, 50)	95%	50%	+45%	98%	100%	−2%
(7, 75)	93%	31%	+62%	100%	100%	+0%
(10, 30)	99%	46%	+53%	100%	100%	+0%
(10, 50)	93%	32%	+61%	100%	98%	+2%
(10, 75)	92%	14%	+78%	100%	98%	+2%
(15, 15)	95%	44%	+51%	100%	100%	+0%
(15, 20)	92%	32%	+60%	100%	100%	+0%
(15, 30)	92%	22%	+70%	100%	100%	+0%
(15, 50)	85%	12%	+73%	100%	98%	+2%
(15, 75)	82%	9%	+73%	100%	100%	+0%
(20, 15)	95%	32%	+63%	98%	100%	−2%
(20, 20)	89%	22%	+67%	100%	100%	+0%
(20, 30)	83%	16%	+67%	100%	98%	+2%
(20, 50)	85%	8%	+77%	100%	92%	+8%
(20, 75)	85%	7%	+78%	100%	90%	+10%
(25, 10)	95%	32%	+63%	100%	100%	+0%
(25, 15)	88%	13%	+75%	100%	100%	+0%
(25, 20)	80%	12%	+68%	100%	93%	+7%
(25, 30)	86%	4%	+82%	98%	100%	−2%
(25, 50)	81%	9%	+72%	99%	91%	+8%
(25, 75)	—	—	—	99%	89%	+10%
(30, 10)	94%	21%	+73%	100%	100%	+0%
(30, 15)	83%	16%	+67%	100%	100%	+0%
(30, 20)	89%	9%	+80%	100%	98%	+2%
(30, 30)	91%	4%	+87%	100%	90%	+10%
(30, 50)	87%	6%	+81%	97%	93%	+4%
(30, 75)	—	—	—	96%	88%	+8%

Table 10: Gemma-3-4b-it LoRA, full held-out Arbitrary-Single results. All 28 held-out cells; FVQ and CVQ accuracy with the gap before and after the adapter. — indicates a baseline cell where the trial count was insufficient; post-adapter result is unaffected.

Attention routing (Table 18). We replicate the §7.2 attention-routing analysis on Gemma using an HF-direct extractor (TransformerLens exceeded the available GPU memory budget on Gemma-3-4b). Operating point $K=2$, $N=30$, 50 trials, expA_CVQ. The top-15 promoter heads on Gemma span L14–L32 (19 of 34 layers), with Δ in $\Pr(\text{attend to } v_{\text{last}})$ ranging from +0.33 to +0.47. Per-layer mean Δ peaks at L25 (+0.31), with secondary peaks at L20 (+0.26) and L30 (+0.24).

Both architectures recover the FVQ–CVQ task via late-layer attention reweighting; the spatial distribution differs (Qwen concentrated, Gemma distributed), but the functional pattern — promotion of v_{last} at late layers — is shared. Paired Stage-3 ablation on Gemma is left to future work; the probing, logit-lens, and attention-routing convergences here are the cross-family evidence we report.

(K, N)	Base			+ LoRA		
	FVQ	CVQ	gap	FVQ	CVQ	gap
(7, 10)	95%	83%	+12%	100%	100%	+0%
(7, 20)	98%	81%	+17%	100%	96%	+4%
(7, 30)	95%	68%	+27%	100%	100%	+0%
(7, 50)	87%	56%	+31%	100%	98%	+2%
(10, 10)	98%	80%	+18%	100%	100%	+0%
(10, 20)	92%	71%	+21%	100%	100%	+0%
(10, 30)	94%	60%	+34%	100%	100%	+0%
(10, 50)	86%	45%	+41%	100%	92%	+7%
(15, 10)	98%	66%	+32%	100%	100%	+0%
(15, 20)	95%	53%	+42%	100%	100%	+0%
(15, 30)	84%	37%	+47%	100%	96%	+4%
(15, 50)	72%	25%	+47%	100%	96%	+4%
(20, 10)	95%	48%	+47%	100%	100%	+0%
(20, 20)	87%	41%	+46%	100%	96%	+4%
(20, 30)	94%	21%	+73%	100%	91%	+9%
(20, 50)	78%	23%	+55%	100%	83%	+17%

Table 11: Gemma-3-4b-it LoRA, Semantic-Multi out-of-distribution results. Adapter trained only on Arbitrary-Single; 16 SEM cells share (K, N) shape with training but use multi-token semantic values. LoRA holds across this distribution shift.

Property	Qwen2.5-3B-Inst	Gemma-3-4b-it
Layers $\times$ heads	36L $\times$ 16H, dense	34L $\times$ 8H, GQA
Baseline regime	reversal (≤ -55 pp)	primacy ($\leq +87$ pp)
ARB held-out fixed	16 / 16	28 / 28
SEM OOD fixed	4 / 4	16 / 16
Trainable params	7.4M (0.24%)	8.9M (0.23%)
Training epochs	2.0	1.42 (early-stop)

Table 12: Cross-family LoRA recipe summary. Same recipe, opposite baseline regimes, near-identical post-LoRA accuracy.

J Block format and LoRA recruit different internal circuits

§5.2 reports that format intervention and LoRA both close the FVQ–CVQ gap behaviourally. We test whether the two interventions share an internal locus on Qwen2.5-3B-Instruct.

Experimental design. Three conditions at $K=2$, $N=30$, 50 trials each (expA_CVQ): (A) base + Plain format, (B) base + Block format, (C) base + main LoRA + Plain format. For each of the 15 LoRA-promoter heads in L30–L33 identified in §7.2, we compute $\Pr(\text{attend to } v_{\text{last}})$ at the generation position.

Behavioural pre-check. At $K=2$, $N=30$ on Qwen, Block lifts CVQ from 49% to 74% (FVQ 89% $\rightarrow$ 98%; gap +40 pp $\rightarrow$ +24 pp; $n = 100$): Block produces a substantial behavioural effect at this cell, so the mechanism question is well posed.

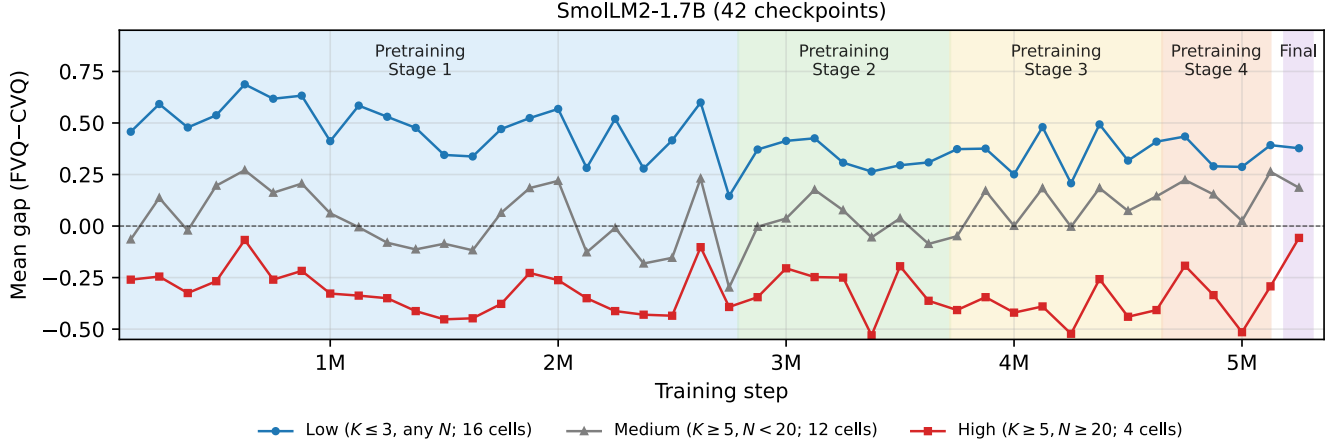


Figure 7: Mean FVQ–CVQ gap on Arbitrary-Single across the SmoLM2-1.7B (Ben Allal et al. 2025) training trajectory, under the same three-bin partition as Figure 4. Coloured bands mark the four documented pretraining stages plus the released final model.

Pooled result (Table 19).

Per-head verdict. Out of 15 LoRA-promoter heads, **0 reach LoRA-level activation under Block** (pre-registered criterion: $\Pr_B$ within bootstrap CI of $\Pr_C$ AND $\Pr_B > 0.4$). 4 of 15 are statistically indistinguishable from baseline under Block (L31H15, L32H10, L32H14, L33H14). The remaining 11 partially activate but stay well below LoRA. No head reaches LoRA’s activation level.

Verdict. Block format and LoRA produce the same behavioural outcome via different internal routes. Block presumably exploits the explicit [Round j] headers as a discrete-step positional cue; LoRA installs attention reweighting in the L30–L33 cluster. The relationship is operational distinctness, not shared mechanism.

K MLP-only LoRA control

§5.2 fits LoRA on the four attention projections $\{Q, K, V, O\}$. A skeptical reading would observe that attention-only LoRA *can only* reweight attention, so “amplification” findings are partly structural. To probe this, we train a control LoRA targeting only MLP modules ($\{\text{gate, up, down}\}_{\text{proj}}$) on the same data.

Training. Identical configuration to §5.2: rank 16, $\alpha=32$, dropout 0.05, lr 2×10^{-4} cosine, effective batch 64, bf16, on Qwen2.5-3B-Instruct. Capped at 100 optimisation steps (the attention-only LoRA took ~ 564 steps to reach comparable eval accuracy). Trainable parameters: 22.6M (0.73% of model weights), $\sim 1.5\times$ the attention-only count because $d_{\text{intermediate}} > d_{\text{model}}$ in Qwen2.5-3B. Eval accuracy reaches 97.1% at step 100.

5-cell preview (Table 20). We evaluated on the five hardest cells in the held-out grid ($n = 100$ trials/cell). All five reach $\text{FVQ} \geq 96\%$ and $\text{CVQ} \geq 94\%$ — indistinguishable from attention-only LoRA within trial-level noise. Full 28-cell evaluation and Semantic-Multi OOD test are open.

Step	FVQ	CVQ	Gap	Cells
125,000	0.37	0.20	+0.17	32
250,000	0.50	0.18	+0.32	32
375,000	0.49	0.30	+0.19	32
500,000	0.56	0.25	+0.31	32
625,000	0.58	0.14	+0.44	32
750,000	0.58	0.24	+0.34	32
875,000	0.59	0.22	+0.37	32
1,000,000	0.55	0.37	+0.19	32
1,125,000	0.57	0.32	+0.25	32
1,250,000	0.52	0.33	+0.19	32
1,375,000	0.50	0.36	+0.14	32
1,500,000	0.53	0.45	+0.08	32
1,625,000	0.48	0.41	+0.07	32
1,750,000	0.52	0.31	+0.21	32
1,875,000	0.56	0.26	+0.30	32
2,000,000	0.57	0.23	+0.33	32
2,125,000	0.40	0.35	+0.05	32
2,250,000	0.55	0.34	+0.20	32
2,375,000	0.46	0.44	+0.02	32
2,500,000	0.52	0.43	+0.10	32
2,625,000	0.57	0.19	+0.37	32
2,750,000	0.38	0.47	−0.09	32
2,875,000	0.51	0.37	+0.14	32
3,000,000	0.53	0.33	+0.19	32
3,125,000	0.61	0.36	+0.25	32
3,250,000	0.59	0.44	+0.15	32
3,375,000	0.55	0.51	+0.05	32
3,500,000	0.58	0.44	+0.14	32
3,625,000	0.56	0.48	+0.08	32
3,750,000	0.58	0.46	+0.12	32
3,875,000	0.66	0.45	+0.21	32
4,000,000	0.58	0.51	+0.07	32
4,125,000	0.65	0.39	+0.26	32
4,250,000	0.63	0.59	+0.04	32
4,375,000	0.62	0.34	+0.28	32
4,500,000	0.60	0.47	+0.13	32
4,625,000	0.62	0.41	+0.21	32
4,750,000	0.70	0.43	+0.28	32
4,875,000	0.67	0.50	+0.16	32
5,000,000	0.64	0.55	+0.09	32
5,125,000	0.73	0.47	+0.26	32
final	0.75	0.50	+0.25	32

Table 13: SmolLM2-1.7B per-checkpoint mean FVQ, CVQ, gap on Arbitrary-Single. 32 cells per checkpoint; 100 trials/cell.

Checkpoint	FVQ	CVQ	Gap	Cells
<i>Pretraining (completion_few_shot)</i>				
stage1-step-40,000	0.66	0.07	+0.59	32
stage1-step-160,000	0.74	0.40	+0.34	32
stage1-step-320,000	0.86	0.33	+0.53	32
stage1-step-440,000	0.88	0.43	+0.45	32
stage1-step-600,000	0.81	0.42	+0.40	32
stage1-step-1,160,000	0.76	0.33	+0.42	32
stage1-step-1,280,000	0.72	0.45	+0.27	32
stage1-step-1,440,000	0.83	0.43	+0.40	32
stage1-step-1,560,000	0.79	0.45	+0.33	32
stage1-step-1,720,000	0.72	0.33	+0.39	32
stage1-step-1,880,000	0.62	0.35	+0.27	32
stage1-step-2,000,000	0.69	0.51	+0.18	32
stage1-step-2,160,000	0.61	0.55	+0.07	32
stage1-step-2,280,000	0.72	0.40	+0.32	32
stage1-step-2,440,000	0.66	0.60	+0.06	32
stage1-step-2,560,000	0.76	0.46	+0.30	32
stage1-step-2,720,000	0.79	0.31	+0.48	32
stage1-step-2,840,000	0.85	0.30	+0.55	32
stage1-step-3,000,000	0.79	0.51	+0.28	32
stage1-step-3,120,000	0.77	0.59	+0.17	32
stage1-step-3,280,000	0.68	0.59	+0.09	32
stage1-step-3,440,000	0.70	0.57	+0.13	32
stage2-step-3,480,000	0.75	0.43	+0.32	32
stage2-step-3,640,000	0.75	0.53	+0.22	32
stage2-step-3,840,000	0.75	0.57	+0.18	32
stage2-step-4,000,000	0.82	0.71	+0.11	32
stage2-step-4,200,000	0.80	0.55	+0.25	32
stage3-step-4,240,000	0.83	0.48	+0.35	32
stage3-step-4,360,000	0.82	0.49	+0.32	32
stage3-step-4,480,000	0.80	0.44	+0.36	32
stage3-step-4,600,000	0.82	0.49	+0.33	32
stage3-step-4,720,000	0.80	0.58	+0.22	32
<i>Post-training (completion)</i>				
it-mid-training	0.79	0.53	+0.25	32
it-SFT	0.82	0.63	+0.19	32
it-soup-APO	0.80	0.61	+0.19	32
it-LC-expert	0.83	0.55	+0.27	32
final	0.81	0.64	+0.16	32

Table 14: SmolLM3-3B per-checkpoint mean FVQ, CVQ, gap on Arbitrary-Single (completion format throughout). 32 cells per checkpoint; 100 trials/cell.

Stage	Completion		Chat (1024)	
	Gap	Cells	Gap	Cells
it-mid-training	+0.25	32	−0.00	32
it-SFT	+0.19	32	+0.29	32
it-soup-APO	+0.19	32	+0.20	32
it-LC-expert	+0.27	32	+0.33	32
final	+0.16	32	+0.18	32

Table 15: SmolLM3-3B post-training stages: mean gap under completion vs chat_1024 prompts. The chat template returns near-zero accuracy on it-mid-training; the gap stays positive under both formats elsewhere.

Layer	Δ Cond. discr.	Δ FVQ correctness	Δ CVQ correctness
L26	+0.00 [+0.00, +0.00]	−0.09 [−0.18, +0.01]	+0.11 [+0.01, +0.21]
L27	+0.00 [+0.00, +0.00]	−0.09 [−0.18, +0.01]	+0.15 [+0.04, +0.26]
L28	+0.00 [+0.00, +0.00]	−0.02 [−0.11, +0.07]	+0.15 [+0.04, +0.26]
L29	+0.00 [+0.00, +0.00]	−0.01 [−0.09, +0.07]	+0.18 [+0.10, +0.27]
L30	+0.00 [+0.00, +0.00]	+0.00 [−0.08, +0.09]	+0.18 [+0.11, +0.25]
L31	+0.00 [+0.00, +0.00]	−0.02 [−0.10, +0.06]	+0.17 [+0.10, +0.24]
L32	+0.00 [+0.00, +0.00]	+0.00 [−0.07, +0.07]	+0.14 [+0.07, +0.21]
L33	+0.00 [−0.00, +0.01]	+0.02 [−0.08, +0.12]	+0.15 [+0.02, +0.28]

Table 16: Gemma-3-4b-it: Δ (LoRA — base) in probing accuracy across late layers, with 95% CIs. Bold entries exclude zero. The condition-discrimination probe saturates pre-LoRA in both architectures. FVQ-correctness is broadly unchanged. CVQ-correctness Δ is positive and statistically distinguishable from zero at all eight late layers, mirroring the Qwen result of Table 5.

(K, N)	$P(v_{\text{first}})$ (FVQ)			$P(v_{\text{last}})$ (CVQ)		
	Base	+LoRA	Δ	Base	+LoRA	Δ
(2, 5)	0.98 [0.95, 1.00]	1.00 [1.00, 1.00]	+0.02 [+0.00, +0.05]	0.80 [0.72, 0.87]	1.00 [1.00, 1.00]	+0.20 [+0.12, +0.28]
(2, 10)	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]	+0.00 [−0.00, +0.00]	0.56 [0.47, 0.65]	1.00 [1.00, 1.00]	+0.44 [+0.34, +0.53]
(2, 50)	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]	+0.00 [+0.00, +0.00]	0.59 [0.50, 0.68]	0.99 [0.99, 1.00]	+0.41 [+0.31, +0.49]

Table 17: Gemma-3-4b-it logit lens at L33 (final layer), baseline vs LoRA, three cells. Bold Δ entries exclude zero. The pre-LoRA $\Pr(v_{\text{last}})$ at the final layer is 0.56–0.80 across these cells (compared with 0.11 in Qwen at $K=2, N=5$); the post-LoRA value is 0.99–1.00 across all three cells. The starting points differ between architectures, but the post-LoRA endpoints coincide.

Property	Qwen2.5-3B-Inst	Gemma-3-4b-it
Layers $\times$ heads	36L $\times$ 16H	34L $\times$ 8H (GQA)
Top-15 promoter range	L30–L33 (4L)	L14–L32 (19L)
Max single-head Δ	+0.68 (L32H3)	+0.47 (L25H0)
Baseline suppressor cluster	L26–L30 (late)	L17–L19 + L30H6
Topology	concentrated	distributed

Table 18: Cross-architecture attention-routing comparison. Both architectures recover via late-layer attention reweighting toward v_{last} . Qwen’s promoters concentrate in the last four layers; Gemma’s spread across mid-to-late depth with three local peaks. The promoter topology in each architecture sits close to its baseline suppressor cluster.

Condition	Pooled $\Pr(\text{attend})$
A: base + Plain	0.103
B: base + Block	0.160
C: base + LoRA + Plain	0.676
$\Delta(B - A)$	+0.056
$\Delta(C - A)$	+0.573
$\Delta(B - C)$	−0.516

Table 19: Pooled $\Pr(\text{attend to } v_{\text{last}})$ across the 15 LoRA-promoter heads in L30–L33. Block format barely activates the cluster (+0.056 above baseline); LoRA strongly activates it (+0.573). Block falls 51 pp short of LoRA’s activation despite producing the same behavioural recovery.

(K, N)	Base FVQ / CVQ	+MLP-LoRA FVQ / CVQ
(10, 50)	42% / 50%	98% / 96%
(15, 20)	32% / 58%	100% / 98%
(20, 30)	21% / 51%	96% / 96%
(25, 75)	6% / 60%	100% / 94%
(30, 75)	1% / 56%	96% / 96%

Table 20: MLP-only LoRA on Qwen2.5-3B-Instruct, 5-cell preview at the hardest part of the held-out grid (100 trials/cell). All five cells close to ceiling; comparable to attention-only LoRA on the same cells (cf. Table 3).

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